



A set of efficient heuristics for a home healthcare problem

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Abstract

Nowadays, the aging population and the little availability of informal care are two of the several factors leading to an increased need for assisted living support. Hence, home healthcare (HHC) operations including a set of nurses and patients have been developed recently by both academia and health practitioners to consider elderlies' preferences willing to receive their cares at their homes instead of hospitals or retirement homes. Commonly, different services, e.g., nursing, physiotherapy, housekeeping and cleaning, for an HHC system are performed by nurses at patients' homes after scheduling and routing the nurses by decision makers. Due to the difficulty of the problem, recent studies show a great deal of interest in applying various metaheuristics and heuristics to solve this problem. To alleviate the drawbacks of previous works and make HHC more practical, this paper develops not only a new mathematical formulation considering new suppositions in this research area but also a lower bound based on Lagrangian relaxation theory has been employed. As such, three new heuristics and a hybrid constructive metaheuristic are utilized in this study to solve the proposed model. Finally, the performance of the proposed algorithms is validated by the developed lower bound and also analyzed by different criteria and also the efficiency of developed formulation is probed through some sensitivity analyses.

Keywords Home health care · Lagrangian relaxation theory · Heuristics · Hybrid constructive metaheuristic

1 Introduction

Nowadays, an increase in the life expectancy for developed countries, e.g., Sweden, Germany, Austria and the USA, as well as a decrease in the preference of old people for the informal care likely led to an increase for the demand of

home health care (HHC)'s services, especially in the last decade [1]. These services are provided by chronic disease at patients' homes [2]. Generally, the HHC has been referred to several names such as in-home care, social care, or domiciliary care [3]. Conditions requiring the HHC most frequently include diabetes, heart failure and chronic ulcer of the skin along with osteoarthritis and also hypertension [4].

Notably, there are numerous reasons to confirm that the home care industry is seeing such accelerating growth. Millions of people are reaching an age where they will need some sort of home care [5]. They are also realizing that home care is a much more affordable way for providing people with the assistance they need [6]. This is true especially when compared with such things as assisted living facilities and nursing homes. People are also beginning to understand the importance of helping the elderly be able to age in place or stay in their own home [3]. This helps them maintain some independence, be happier and be able to stay in their familiar surroundings. Overall, a home care is the option that allows them to do this [2].

According to the latest statistics available from the National Association for Home Care and Hospice,

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approximately 12 million people in the USA require some form of home health care. Also, more than 33,000 home healthcare providers exist today and almost two-thirds (63.8 percent) of home healthcare recipients are women [2]. More than two-thirds (69.1 percent) of HHC recipients are over age 65. In the USA, Medicare is the largest single payer of home care services. For instance, Medicare spending was approximately 41 percent of the total home healthcare and hospice expenditure in 2009 [7]. As such, approximately \$72.2 billion was projected to be spent on home health care in this year. Accordingly, the number of persons employed in the US HHC is surprisingly increased during this century. Figure 1 shows a statistical graph related to this issue.¹ As indicated from this report, there were approximately 1,495,000 persons employed in HHC of USA in 2016. Clearly, this issue reveals the importance of economic driver of HHC services.

The main responsibility of companies supporting the HHC services is to provide the required drugs and biological samples [3]. In addition, the nurses are routed to perform different HHC services, and then they are scheduled to set the time table of patients for providing the patients required due to the degree of recovery [8]. In a nutshell, the main operations performed by such companies [2, 5–7, 9] can be summarized as follows:

- (i) Firstly, the whole necessary data from registered patients should be collected by the companies. The information includes the name, gender, age, address, types of diseases, history of their illness, symptoms, expected service time and other necessary data for the patients.
- (ii) Secondly, a plan is required to design the assigned route for each patient. Nurses or doctors have to start from a pharmacy and after visiting the assigned patients of this route, they should transfer the biological samples of patients to a laboratory to analyze them. Accordingly, a performance plan of assigned route determines the provision of HHC services involving of delivering the right quantity of drugs as well as assisting the patients within their specified time horizon.
- (iii) Finally, the rostering and scheduling of nurses or doctors to visit the patients according to the previous step should be provided. In each route, a nurse or a group of nurses may be chosen to visit the assigned patients. As such, the demand of each patient should be supplied. To do this end, the transforming of nurses by using different kinds of vehicles is needed to be explored.

The transportation cost is one of the main economic aspects of HHC companies [1]. Several recent studies such as [2, 3, 5, 10, 11] have tackled this issue using an optimization model to reduce the transportation costs of HHC.

In order to control the transportation cost, this paper aims to consider a policy for overall distance travelled for assigned patients by considering different types of vehicles to transform the nurses [8]. The effect of this rule will be studied in this study, properly. In addition to this new development, a Lagrangian relaxation-based algorithm is employed to find a feasible lower bound for the developed model. Another contribution is to introduce three fast heuristics along with a hybrid constructive metaheuristic. The main contributions are summarized as follows:

- A new mixed integer linear programming formulation for a home healthcare problem is proposed.
- A lower bound based on Lagrangian relaxation methodology is developed for the problem.
- Three fast heuristic algorithms as well as a hybrid constructive metaheuristic are proposed to solve the problem.
- Several experiments with different complexity's levels and some sensitive analyses have been done.

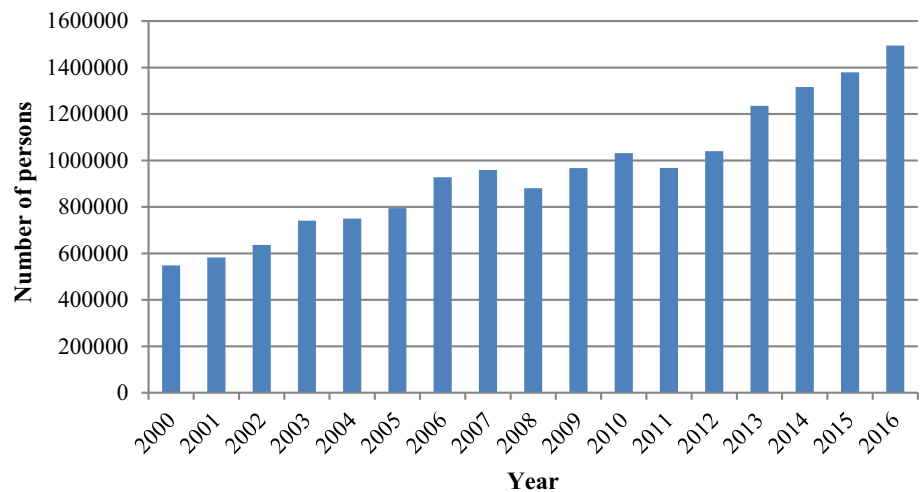
The rest of the paper is organized as follows. Section 2 reviews the related literature mainly on the model's characteristics and solution approach. In Sect. 3, the proposed mathematical formulation is illustrated and detailed. In addition to this new development, it includes the solution methodology as the main innovation of this study. Afterward, the computational analyses by considering different criteria based on the solution time and quality and sensitivity analyses for key parameters of model are discussed in Sect. 4. Finally, the conclusion and future directions of this study are presented in Sect. 5.

2 Literature review

The home healthcare problem (HHCP) is a preliminary of vehicle routing problem (VRP) [3] which considers a company willing to supply the clients' demands using a number of vehicles [2]. The goal of VRP is to optimize the total cost of traveling for vehicles. As a type of combinatorial optimization problems with several real-life constraints, it is known as a NP-hard problem and its complexity is directly related to the number of clients and vehicles [7]. Exact algorithms, e.g., Branch & Bound or Branch & Cut, are often not able to find optimal solution for this problem in a reasonable time [11]. This is the main reason for the use of heuristics and metaheuristics as the popular feasible alternatives for solving such NP-hard problems [3].

¹ <https://www.statista.com/statistics/185249/persons-employed-in-home-health-care-services-in-the-us-since-2000/>.

Fig. 1 The number of persons employed in HHC services in the USA from 2000 to 2016



Due to the novelty of HHCP, there are a few review papers in this area [1]. It is reported that the literature of HHCP includes fifty papers in two recent decades [1, 5]. Most of the mathematical models in this research area are a type of mixed integer linear programming (MILP). One of the first studies can be referred to Begur et al. [12], in which they developed a spatial decision support system (SDSS) in a case study taken from the USA. Additionally, Cheng and Rich [13] proposed a new MILP formulation firstly by considering a soft form of time window in this area. They also proposed a simple heuristic to address the model. In 2006, Bertels and Fahle [14] proposed a hyper-heuristic algorithm by combining a linear programming, constraint programming and some heuristic techniques employed by [13]. They classified their model as a general type of VRP. Moreover, another decision technique was addressed by Eveborn et al. [15]. They applied a decision support system (DSS) by adding L_{APS} and C_{ARE} to control the decision policy through a real case study in Sweden.

As a type of combinatorial optimization, metaheuristics or generally nature-inspired optimizers have been employed repeatedly to solve the HHCPs especially in the recent decade. For instance, Akjiratikarl et al. [16] proposed the use of a particle swarm optimization (PSO) in solving an HHCP through a case study in Ukraine. The main contribution of their work was to apply a heuristic technique named earliest start time priority with minimum distance assignment (ESTPDA). Furthermore, Trautsamwieser et al. [17] introduced a variable neighborhood search (VNS) based on heuristic procedures to solve an HHCP under uncertainty for natural disasters inspired by a flood one in 2002 from Austria. In another work, by a case study in Sweden, some preliminary mathematical formulas were provided to highlight the role of operation research (OR) models to find an optimized HHCP by Eveborn et al.

[18]. They also indicated that metaheuristics are effective approaches to solve the underlying HHCP issues.

In 2009, a hard form of time window in this research was firstly proposed by Kergosien et al. [19]. They also developed a fast heuristic to reduce the computational time in large sizes of developed HHCP. In a similar research, a home care crew scheduling problem as a type of VRP with time window (VRPTW) was suggested by Rasmussen et al. [20]. To solve their mathematical model, they proposed an exact branch-and-price solution planning. In addition to this development, master schedule problem (MSP) and operational planning problem (OPP) were firstly added into the HHCP by Nickel et al. [21]. The test problems are generated according to a real case study in Germany. As such, they introduced a heuristic algorithm based on the constraint programming approach. In another recent study, Liu et al. [6] considered different heuristics based on the genetic algorithm (GA) and Tabu search (TS) to address the HHCP with the delivery time and time window suppositions. In 2014, Liu et al. [7] developed a hybrid TS based on local searches to explore the feasible and infeasible solutions by using benchmark instances in VRP.

There is a great deal of attention in considering new real assumptions for HHCPs in recent years. For example, Mankowska et al. [22] added the interdependencies of services in a new HHCP. Their considered services include several items, e.g., a temporal separation of two services as is needed if drugs have to be administered a certain time before providing a meal or for handling a disabled patient, perhaps, he or she requires two nurses working together at the same time in patient's home. Some sensitivity analyses for a new MILP model by the use of exact method implemented in GAMS software were done. In 2015, Hiermann et al. [23] offered a multi-modal HHCP by considering a case study in Austria. The main contribution was to propose a two-stage solution methodology. The first

stage was the constraint programming approach to reduce the constraints and to expand the feasible space. The second stage was to employ a number of metaheuristics to find the optimal solution of problem. The metaheuristics employed include VNS, memetic algorithm (MA), scatter search (SS) and a simulated annealing hyperheuristic. In another recent paper, Fikar and Hirsch [8] added some new assumptions including the possibility of walking to patients, the interdependencies of services, the time window as well as the mandatory working time and the break regulations; whole of them by using different transportation systems. They generated the test problems in both small and large sizes by supporting Austrian Red Cross as a case study. At the last but not the least, a two-stage hybrid metaheuristic was developed to tackle their NP-hard problem.

Although most of the studies consider a single-objective form of MILP, the study of Braekers et al. [10] was among the first studies to develop a bi-objective HHCP. Their goal was to find a trade-off between the total cost and patient inconvenience by using a metaheuristic based on local searches. In 2017, Shi et al. [2] introduced VRPTW based on fuzzy numbers. They applied a hybrid GA to solve the benchmarked instances of VRPTW. Contrary to the mentioned literature, Cappanera et al. [4] developed a robust optimization model to formulate an HHCP under demand uncertainty in a multiple-day horizon time. They addressed their model by a non-standard cardinality-constraint robust approach as well. In 2018, Lin et al. [3] developed an integrated model to consider both nurse scheduling and VRPTW. As a complicated optimization problem, the authors used a hybrid of harmony search algorithm (HSA) and GA to estimate the optimal solution for the model. More recently, a green HHCP was developed by Fathollahi-Fard et al. [5]. They proposed different heuristics and metaheuristics to solve their model. The aim of their work was to find a trade-off between the transportation costs and environmental pollution in an HHCP.

Taken together, the current works can be divided into three groups: improving model, improving current optimizers or proposing new optimizers and improving both model and optimizer. HHCP can be formulated and modeled with different methods and considering more factors. Although adding more elements makes the HHCP more practical, it increases the difficulty and computation cost of this problem. Additionally, according to the no free lunch theorem [24], there is no optimization algorithm to solve all optimization problems. This means that changing case model and/or case study will change the search space of the problem and might result in getting poor results using the current heuristic methods [25, 26]. These facts keep this area active and motivate our attempts to contribute to both areas: model and optimizer improvement.

According to the literature along with the best of our knowledge, this study is among the first studies to consider a penalty for overall distance traveled by assuming the type of transportation systems as well as the time window and the delivery time of patients. Moreover, the main innovations of this study are to develop a lower bound based on Lagrangian relaxation theory for the first time in the literature as well as three quick heuristics and a hybrid metaheuristic algorithm. In the next section, the proposed methodology to formulate and to solve this problem has been ordered in detail.

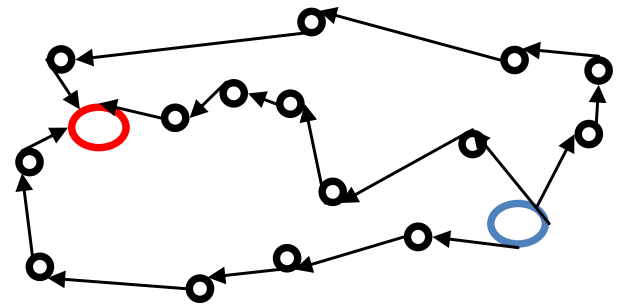
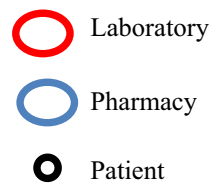
3 Proposed methodology

This section not only proposes a new MILP formulation an HHCP but also a lower bound, and a set of efficient heuristics and hybrid algorithm are developed to tackle this problem, comprehensively. The main decision variables of HHCP are scheduling and routing of nurses to visit the patients with different difficulties, e.g., the patient's availability and favorability, the budget of company, cost and capacity of vehicles and controlling the routes to decrease the transportation cost. Commonly, each nurse or doctor is assigned to only one route. The main activities of a HHC company reveal that the drugs or medical instruments from a pharmacy to the patients should be delivered. Then, after visiting all patients, the biological samples are collected to transform from patients' home to the laboratory. To graph the proposed HHCP, Fig. 2 depicts an assumed test problem with 15 patients and three nurses.

3.1 Problem description

Generally, the description of developed HHCP inspired by Shi et al. [2] can be addressed as follows. First of all, consider a city or a town with M patients distributed in this geographical zone. A company by establishing only one pharmacy and one laboratory aims to supply the patients demand and to perform the HHC services. It is estimated that each patient requires a specific amount of drugs equaled to A_i . Similarly, the biological samples of patients (B_i) should be collected and transformed to a laboratory. Furthermore, the company employs N nurses to address the demand. Each of them is allocated to one route which starts from pharmacy to laboratory. In the proposed HHCP, the distances are considered in a two-dimensional geographical space. In this regard, D_{ij} is the distance between patient i and patient j . According to the time window's constraints, each patient has the earliest time (E_i) and the latest time (L_i) to visit by nurses. In regard to HHC services, e.g., nursing, physiotherapy, housekeeping and cleaning, a

Fig. 2 Graphical illustration of proposed HHCP [2]



specified time for each patient (W_i) is considered approximately. The nurses for transforming from the patient i to the patient j need a specified time (T_{ij}) according to the used transportation systems. As discussed earlier, each nurse starts from his/her pharmacy to visit the patients and go back to his/her laboratory. In each route, one vehicle should be considered. This paper considers different types of transportation systems, e.g., car, public and so on. Generally, there are K vehicles in each of them has a capacity of CAP_k and transportation cost of TC_k . One of the main contributions of the model is *to consider a penalty for overall distance*. It is obvious that the tours of patients should be generated in a way to make a balance between the distances traveling of each tour and a desirable amount of cost. On the other hand, the summation of distance traveled for the nurses and their employed vehicles should be lower than a predefined number ($MDIS_{nk}$). If this policy is not observed, a penalty value will be considered for the extra traveled distance (PEN). In conclusion, the aim of this innovation is to better reach an optimized solution to cover more economic aspects of HHCP under study.

3.2 Assumptions

The HHCP developed considers the following assumptions:

- The developed model is a type of single-objective optimization MILP model by considering only one objective and a single period and depot.
- The HHCP involves numerous patients with a set of nurses and only one pharmacy and one laboratory.
- The demand (required drug) of each patient must be met. There is no difference between the types of products as the required drug of patients.
- There are different types of transportation systems including different cars, public trains, etc., in which they are considered in the model by assigning different types of capacities and transportation costs of the vehicle type k for each unit of traveled distance from patient i to patient j .

- Each nurse starts from the pharmacy and goes to the laboratory after visiting his/her assigned patients.
- The working time and required drugs for each patient should be estimated before the visit.
- The location of the pharmacy and laboratory is predefined.
- The time window is considered for the HHC services according to the availability of patients and calculated by estimating the earliest and the latest service time for the patients.
- To control the transportation cost more efficiently, a penalty value is considered for the overall distance traveled by the nurses to visit their patients according to the policy of company.

3.3 Notations

The notations of developed HHCP including indices, parameters and decision variables are stated as follows:

Indices:

- i, j Indices of patients, $i, j \in \{1, 2, \dots, M\}$
 n Index of nurses, $n \in \{1, 2, \dots, N\}$
 k Index of transport systems, $k \in \{1, 2, \dots, K\}$

Parameters:

- D_{ij} Distance between patient i and patient j
 TC_k Transportation cost per unit of distance for transport system k
 CAP_k The capacity of transport system k
 W_i The working time for the nurse to service the patient i
 E_i The earliest service time for the patient i
 L_i The latest service time for the patient i
 T_{ij} The traveling time from patient i to patient j
PEN Amount of penalty for nurses for over distance traveling between patients ($1 < PEN < 5$)
BIG A positive large number, this parameter is generated to construct the time window constraints
 $MDIS_{nk}$ Maximum desired traveled distance for the nurse n by using transportation system k

- A_i Demand of patient i (amount of drugs required for patient i)
- B_i Biological samples collected from patient i to transform the laboratory

Decision variables:

- X_{ijn}^k 1 if the nurse n by using transport system k visits the patient i before patient j , otherwise 0
- S_{in} The time at which nurse n begins to service the patient i . In this regard, if in an event the given nurse n does not service patient i , S_{in} does not mean anything
- O_{nk} Overall distance traveled by nurse n using transport system k

3.4 Formulation

The MILP formulation that has been proposed in this work for HHCP is explained as follows:

$Z = \text{Min}$

$$\left(\sum_{k=1}^K \sum_{n=1}^N \sum_{i=1}^M \sum_{j=1}^M D_{ij} \times \text{TC}_k \times X_{ijn}^k + \sum_{n=1}^N \sum_{k=1}^K O_{nk} \times \text{TC}_k \times \text{PEN} \right) \quad (1)$$

s.t.

$$\sum_{n=1}^N \sum_{k=1}^K \sum_{j=1}^M X_{ijn}^k = 1, \quad \forall i \in M \quad (2)$$

$$\sum_{k=1}^K X_{ijn}^k = 1, \quad \forall i, j \in M, \quad n \in N \quad (3)$$

$$\sum_{i=1}^M A_i \times \sum_{j=1}^M X_{ijn}^k \leq \text{CAP}_k, \quad \forall k \in K, \quad n \in N \quad (4)$$

$$\sum_{i=1}^M B_i \times \sum_{j=1}^M X_{ijn}^k \leq \text{CAP}_k, \quad \forall k \in K, \quad n \in N \quad (5)$$

$$\sum_{k=1}^K \sum_{j=1}^M X_{ojn}^k = 1, \quad \forall n \in N \quad (6)$$

$$\sum_{i=1}^M X_{ihn}^k - \sum_{j=1}^M X_{hjn}^k = 0, \quad \forall h \in M, \quad k \in K, \quad n \in N \quad (7)$$

$$\sum_{i=1}^M X_{i0n}^k = 1, \quad \forall k \in K, \quad n \in N \quad (8)$$

$$S_{in} + T_{ij} + W_i - \text{BIG} \times \left(1 - \sum_{k=1}^K X_{ijn}^k \right) \leq S_{jn}, \quad \forall i, j \in M, k \in K, n \in N \quad (9)$$

$$E_i \leq S_{in} \leq L_i, \quad \forall i \in M, \quad n \in N \quad (10)$$

$$O_{nk} \geq \left(\sum_{i=1}^M \sum_{j=1}^M D_{ij} \times X_{ijn}^k \right) - \text{MDIS}_{nk} \quad (11)$$

$$S_{in}, O_{nk} \geq 0 \quad (12)$$

$$X_{ijn}^k \in \{0, 1\} \quad (13)$$

Similar to the most of HHCPs as a variant of VRP [2, 5, 11], it is typical to calculate the total transportation cost. Equation (1) shows the objective function in two terms to optimize the transportation costs of proposed system. The first and second terms are considered the routing and scheduling of nurses to visit the patients. In the first one, due to traveled distance for the nurses by transportation system type k , the transportation cost is considered to be minimized. As per the second term, the overall traveled distance for nurses regarding the maximum desired traveled distance by considering the type of transportation system is computed by adding a penalty value to calculate the extra transportation costs.

There are also a number of constraints in the model proposed. Equation (2) indicates each patient is visited once only. Equation (3) shows that each nurse is assigned to only one vehicle. Equations (4) and (5) ensure that the used transport system should have enough capacity to transform the patients' drugs as well as the biological samples. Equation (6) represents that each nurse starts from its pharmacy. Equation (7) shows that the nurse visits the patient and then leaves this patient. As indicated from this constraint, it reduces the extra number of infeasible sub-tours for the model. Equation (8) also guarantees that each nurse should go back to the laboratory after visiting the patients. Equation (9) is designed to ensure that each nurse n cannot arrive at patient j before, $S_{in} + T_{ij} + W_i$, since the nurse requires the working time W_i , as well as traveling time from patient i to patient j , here BIG is a large scalar. This constraint makes the model harder. Equations (9) and (10) guarantee that the time window is existed according to the earliest and the latest of availability of patients. Equation (11) calculates the over traveled distance for the nurses from a maximum desired distance by considering the employed transport system. Finally, Eqs. (12) and (13) enforce the non-negativity restrictions and binary variables of the developed HHCP, respectively.

According to the literature and to the best of our knowledge, this MILP formulation has not been studied before in the literature. Since the VRP is known as NP-hard, the developed HHCP is NP-hard, too. Finding an optimal solution for this problem is computationally expensive despite all constraints both soft and hard ones as well as the decision variables [8]. To tackle this hindrance, this study attempts to first examine a Lagrangian

relaxation-based algorithm to find a feasible lower bound for the problem. To better solve the model, three fast and reliable heuristics are proposed. A simple stochastic optimizer as a constructive hybrid metaheuristic is employed to address the problem as well.

3.5 Lagrangian relaxation-based algorithm

This paper employs a Lagrangian relaxation-based algorithm as a type of sub-gradient algorithms to solve the combinatorial optimization problems [27]. The proposed methodology involves two main components, i.e., the lower and upper bounds. First of all, the proposed formulation should be relaxed by eliminating a set of constraints from the model and adding them into the objective function [28]. Then, these constraints are multiplied by Lagrange multipliers. In regard to the present relaxed problem, solving it in the case of minimization provides a lower bound for the original formulation [29]. After that this obtained solution should be utilized in the original model by generating a feasibility procedure. This feasible solution forms the upper bound of the original formulation. By employing these lower and upper bounds, the algorithm aims to fill the gap between them and finds a solution which has both optimality and feasibility properties [30].

To better show the proposed algorithm, we present the relaxed problem in the first step. Consequently, a plan to reach both the lower and upper bounds of the problem with explanations in details is presented to clarify how to find them. Finally, the algorithm tries to reach a feasible lower bound for the original minimization formulation.

3.5.1 Relaxed problem

To achieve the relaxed problem, a number of hard constraints should be relaxed from the original formulation. Since the constraint presented in Eq. (9) is a big-M type constraint, it makes the model more difficult. Setting the path of nurses by considering the availability times of patients reduces the space of feasible solutions. This constraint is also a version of Miller–Tucker–Zemlin sub-tour of elimination constraint [31]. Therefore, the relaxed formulation of the problem with Lagrange multiplier π_{ijn} is as follows:

$$Z'' = \text{Min} \left(Z + \sum_{i=1}^M \sum_{j=1}^M \sum_{n=1}^N \pi_{ijn} \times \left(S_{in} - S_{jn} + T_{ij} + W_i - \text{BIG} \times \left(1 - \sum_{k=1}^K X_{ijn}^k \right) \right) \right) \quad (14)$$

s.t.

Equations (2–8), (10–13)

To achieve the initial lower bound, the above formulation has been solved. Then, the decision variables of model (X_{ijn}^k) are employed by the original formulation to find the lower bound of model. Note that although this solution may be not feasible, it will be an optimal solution for the original problem [28].

Additionally, an upper bound to start the main loop of methodology is needed to be explored. A simple approach to get the upper bound is to change the obtained solution from the relaxed model (lower bound) to make it feasible for the original problem [27]. In this case, the number of assigned nurses and patients should be reduced (X_{ijn}^k) to make it feasible. According to the matrix space of decision variable, a set of arrays equaled to 1 is randomly selected to equal zero. This reduction will be continued to make the feasibility matrix of X_{ijn}^k . By solving the restricted decision variable in the original formulation, the initial upper bound is generated. Note that although this solution may not be optimal, it will be a feasible solution for the original problem.

3.5.2 The main loop of methodology

Regarding the adopted lower bound (LB) and upper bound (UB) for the proposed problem, the main loop of proposed Lagrangian relaxation-based algorithm is structured as follows:

Step 0 Initialize the Lagrange multiplier π_{ijn}^0 and set $t = 0$;

Step 1 Let $\pi_{ijn} = \pi_{ijn}^t$ and solve the relaxed problem (Z''). Then, consider the obtained decision variable with the optimal solution of $Z(\pi_{ijn}^t)$ to update the LB as follows:

$$LB^{t+1} = \max(LB^t, Z(\pi_{ijn}^t)) \quad (15)$$

Step 2 According to the optimal solution of $Z(\pi_{ijn}^t)$, if this solution yields an infeasible solution, the number of assigned patients to the nurses will be decreased until the restricted problem provides a feasible solution. Consequently, the UB will be updated as follows:

$$UB^{t+1} = \min(UB^t, Z_{(X_{ijn}^k)}^t) \quad (16)$$

Step 3 The Lagrange multiplier should be updated as follows:

$$\pi_{ijn}^{t+1} = \max \left(\pi_{ijn}^t + \mu^t \times \left(S_{in} - S_{jn} + T_{ij} + W_i - \text{BIG} \times \left(1 - \sum_{k=1}^K X_{ijn}^k \right) \right), 0 \right) \quad (17)$$

where $\mu^t = f^t \times \left| \frac{\text{UB}^t - \text{UB}^{t+1}}{(\text{UB}^{t+1} - \text{LB}^{t+1})^2} \right|$ and being f a number distributed by $U(0, 2)$ in the first iteration and it is decreased during the number of iterations by $f^{t+1} = f^t \times \alpha$ without any improvement. Note that α is reduction rate taken between 0.5 and 1;

Step 4 $t = t + 1$;

Step 5 If a feasible LB reaches or t satisfies the maximum number of iteration (Maxit) then, stop and display LB. Otherwise, go to Step 1;

Generally speaking, the proposed methodology has three input parameters including π_{ijn}^0 , f^0 and α . These parameters should be tuned before running the algorithm to enhance its performance. Among them, the most important parameter of the algorithm is an initial Lagrange multiplier π_{ijn}^0 . Then, based on this Lagrange multiplier, the relaxed problem has been solved. In regard to generated decision variables, if the total cost of original model is greater than LB, accordingly, the LB is updated. Afterward by changing the optimal solution randomly, the restricted formulation has been solved to reach a feasible solution. Accordingly, the UB should be updated if the objective function of obtained feasible solution is lower than the UB. Consequently, the Lagrange multipliers are updated according to presented formulations in Step 3. As the next step, the relaxed problem to achieve a new LB and UB is solved by new Lagrange multipliers. These processes will be continued until a feasible LB or the maximum number of iterations is reached. The tuning and solving of proposed algorithm are provided in the computational results section.

3.6 Heuristic algorithms

Another innovation of this paper is to present a number of heuristic algorithms to approximate the global optimal solution for the model in a reasonable time. Accordingly, three heuristics by considering a distinguished strategy and focusing on the key parameters of model are developed. The literature is very rich in utilizing heuristics to solve the HHCPs (e.g., Bertels and Fahle [14], Kergosien et al. [19], Nickel et al. [21], Liu et al. [6, 7] and also Fathollahi-Fard et al. [5]). The proposed heuristics are clearly differed from previous works. It should be noted that the two strategies (H1 and H2) proposed by Fathollahi-Fard et al. [5] are the main inspiration of our developed heuristics. In addition to these strategies, a new one (H3) is also considered by this

study to have a comprehensive comparison. Due to novelty of proposed model, all these heuristics are design to better address the proposed problem. The heuristics are set to complete the main binary decision variable of model, i.e., X_{ijn}^k , properly. Each of heuristics is to consider a similar matrix close to this decision variable. To maintain on the objective function, the tours of nurses are designed by considering the distance between patients (D_{ij}) and the rate of transportation cost for each vehicle (TC_k). In this way, the following formula is created the main input of heuristics:

$$\text{DTC}_{ijk} = D_{ij} \times TC_k; \forall i, j \in M; k \in K \quad (18)$$

where DTC_{ijk} is the main input of all heuristics. Only the first step of each heuristic is different and the other steps are the same. The main strategy of them is based on the allocation of patients. Overall, the first step of heuristics is stated as follows:

- Heuristic 1 (H1): to set the tour of per nurse, the first patient is to consider the minimum digit of the first row of the matrix presented in Eq. (18). The other patients are selected by minimum digit of this matrix in the other rows.
- Heuristic 2 (H2): for the tour of each nurse, selecting the first patient is started with the average of each row of the matrix presented in Eq. (18). Similar to H1, selecting the other patients of the tour is based on the minimum digit of this matrix in the other rows.
- Heuristic 3 (H3): the first patient of each tour of each nurse is chosen by the minimum digit of the last row of the matrix presented in Eq. (18). Accordingly, the rest of patients of this tour are selected by considering the minimum digit of this matrix in the other rows.

To consider the differences of heuristics, Fig. 3 reveals three routes by considering five patients. As can be seen, according to H1, the route including $\{2 \rightarrow 4 \rightarrow 1 \rightarrow 3 \rightarrow 5\}$ (Fig. 3a) is started with 2 as the nearest patient of the first patient. Meanwhile, the route including $\{3 \rightarrow 2 \rightarrow 4 \rightarrow 1 \rightarrow 5\}$ (Fig. 3b) is formulated by H2. The average of patient is the patient number 3 which is the start point of the tour. The route including $\{4 \rightarrow 2 \rightarrow 3 \rightarrow 1 \rightarrow 5\}$ is based on the H3 (Fig. 3c). The nearest patient close to patient number 5 is patient number 4 which is the start point in this case.

As mentioned earlier, the rest steps of heuristics are the same. These steps also meet the constraints of model. First of all, the capacity of vehicles should be considered, and a route will be set as per this capacity. When the required drugs for the patients are more than the allowable capacity of used vehicle, another type of vehicle should be employed, and the biological samples from the visited

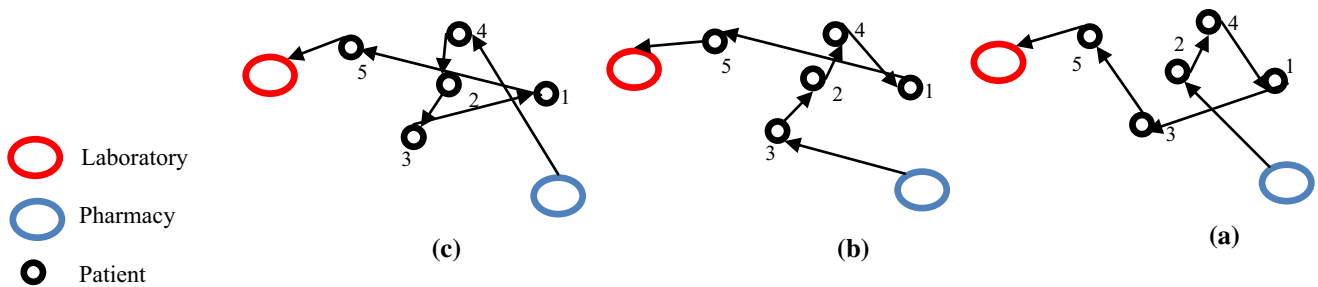


Fig. 3 An example for graphical explanation of effect H1(a) and H2(b) and also H3(c) on the route of a nurse

DTC_{ijk}		1	2	3	4	5	6	7	Average
k=1: two-passenger	1	#	288	231	216	165	186	159	207.5
	2	288	#	111	225	273	198	132	204.5
	3	231	111	#	186	234	90	156	168
	4	216	225	186	#	138	192	195	164.5
	5	165	273	234	138	#	234	108	192
	6	186	198	90	192	234	#	150	175
	7	159	132	156	195	108	150	#	150
k=2: four-passenger	1	#	480	385	360	275	310	265	345.8
	2	480	#	185	375	455	330	220	340.8
	3	385	185	#	310	390	150	260	280
	4	360	375	310	#	230	320	325	274.2
	5	275	455	390	230	#	390	180	320
	6	310	330	150	320	390	#	250	291.6
	7	265	220	260	325	180	250	#	250

Fig. 4 A numerical example of H1, H2 and H3

patients must be transformed to the laboratory. Another main constraint of the model that should be met is the time window. To do this end, during the generation of a route, the route should be changed if a patient is not available. Due to the maximum desired distance of vehicles as a policy of proposed model along with a nurse is available for assigning to a route, the length of this tour should be shortened and the nurse should assigned to the extra patients in an extra tour. To better understand the heuristics, a numerical example for each heuristic is given with illustrative details in Fig. 4.

From Fig. 4, seven patients with two nurses along with two types of vehicles are existed. The rate of transportation cost for two types of vehicles including the large car four-

H1:

The first nurse:

$$0 \rightarrow 5 \rightarrow 4 \rightarrow 3 \rightarrow 6 \rightarrow 0$$

$$X_{051}^1 = X_{541}^1 = X_{431}^1 = X_{361}^1 = X_{601}^1 = 1$$

The second nurse:

$$0 \rightarrow 2 \rightarrow 7 \rightarrow 1 \rightarrow 0$$

$$X_{022}^1 = X_{272}^1 = X_{712}^1 = X_{102}^1 = 1$$

The other similar variables are equal zero.

H2:

The first nurse:

$$0 \rightarrow 7 \rightarrow 5 \rightarrow 4 \rightarrow 3 \rightarrow 0$$

$$X_{071}^1 = X_{751}^1 = X_{541}^1 = X_{431}^1 = X_{301}^1 = 1$$

The second nurse:

$$0 \rightarrow 6 \rightarrow 1 \rightarrow 2 \rightarrow 0$$

$$X_{062}^1 = X_{612}^1 = X_{122}^1 = X_{202}^1 = 1$$

The other similar variables are equal zero.

H3:

The first nurse:

$$0 \rightarrow 2 \rightarrow 3 \rightarrow 6 \rightarrow 7 \rightarrow 0$$

$$X_{021}^1 = X_{231}^1 = X_{361}^1 = X_{671}^1 = X_{701}^1 = 1$$

The second nurse:

$$0 \rightarrow 4 \rightarrow 5 \rightarrow 1 \rightarrow 0$$

$$X_{042}^1 = X_{452}^1 = X_{512}^1 = X_{102}^1 = 1$$

The other similar variables are equal zero.

passenger and another one with two passenger are five and three units, respectively. Note that the number of passengers has not considered in this case. There are only the name of a small and a medium car. A specific route is selected for each heuristic. To run the proposed H1, among all arrays of DTC_{ijk} matrix, in the first row of each vehicle, the minimum one is selected. This means that the fifth patient for the first type of vehicle is selected. After that, the special procedure of H1 is employed as can be seen in Fig. 4. As such, the fourth patient is selected. After the fourth patient, the column of the fifth patient is deleted and then the third patient is chosen by a same procedure similar to earlier. Due to the details of mentioned constraints of the model, values of the main decision variable of the model

```

A set of neighborhood structures  $N_k$  for  $k=1, 2, \dots, k_{max}$  for shaking.
Initialize the temperature and its reduction rate.
Generate the initial solution.
 $X^*=X$ =the best solution.
while ( $t$ < maximum number of iteration)
     $k=1$ ;
    while ( $k < k_{max}$ )
        Shaking: pick a random solution  $X'$  from the  $k^{th}$  neighborhood  $N_k(X)$  of  $X$ .
         $X''$ =local search ( $X'$ );
        if  $f_{new}$ =the objective function of  $X''$  is better than  $f_{old}$ =the objective of  $X'$ 
             $X^*=X''$ ;
            Continue to search with  $N_i$ ;  $k=1$ ;
        else
             $k=k+1$ ;
            Calculate  $\delta$ ,  $\delta = |f_{old} - f_{new}|$ .
            if  $\text{rand} < \exp(-\delta/T)$ 
                Replace the new solution.
            endif
        endif
    endwhile
    Update the  $X^*$  if there is better solution.
    Update  $T$ .
     $t=t+1$ ;
end while
return  $X^*$ 

```

Fig. 5 The pseudo-code of VNS-SA

(X_{ijn}^k) are shown after the routes in a part of Fig. 4. Similarly, to implement H2, the first patient is started with the seventh patient which has the lowest average as can be seen in the matrix of DTC_{ijk} . It is evident that the other steps of H2 are similar to H1. Moreover, H3 is started with the second patient as the nearest patient to the last patient (patient 7). After the ending of the first route, the second one is started with the fourth patient as the closest patient to the fifth patient as the existing last patient.

3.7 Hybrid constructive metaheuristic

As mentioned earlier, there are several successful optimizers utilized in the literature, e.g., VNS, SA and TS etc. (e.g., [5, 6, 23, 32]). Scholars revealed the performance of these simple algorithms to solve the combinatorial optimization problems during the last decade. Notably, one of the main abilities of them is to reach the global solution in most of cases [33, 34]. Recently, the hybridization and modification of these metaheuristics are done by several researches [35–37]. The aim of a hybrid metaheuristic is to enhance a better interaction between the intensification and diversification phases [11]. For instance, in a recent study, Aliakbarian et al, [38] showed that a hybrid of SA and VNS outperform both algorithms. This study employs a hybrid constructive metaheuristic by combining VNS and SA. In the literature of HHCP, the proposed algorithm has not been used before. Due to the novelty, the following

paragraphs provide the details of proposed hybrid algorithm. We do not discuss the details of SA and VNS, individually, since they are well known. Interested readers are referred to see several studies in this research area [5, 29, 32, 35–37, 39, 40].

Here, we call the proposed hybrid constructive metaheuristic as VNS-SA. In the proposed VNS-SA, VNS plays the main loop of algorithm and SA is a sub-loop. SA has this ability to escape from local optima by an accepting rule. It maintains the intensification properties [35]. In addition to SA, VNS is focused on the diversification phase by considering different procedures of shaking solutions' neighbor [36, 37]. Generally, the proposed algorithm is started with an initial random solution. In each iteration, the neighbor of solution is changed by an approach. If an improvement on the cost of solution has not been seen, SA helps the algorithm to consider the acceptance rule of new solutions. Generally speaking, the framework of VNS-SA is similar to VNS except the acceptance of the worst solutions according to probability of SA [38]. The stopping criterion of proposed hybrid constructive metaheuristic is the maximum number of iteration. For more illustration, a pseudo-code is given in Fig. 5.

To encode the proposed VNS-SA for the developed HHCP, a plan based on *Random-Key* is utilized [41]. This technique is a two-stage procedure to transform an infeasible solution to a feasible one with less computational time [40]. This reason motivated several similar papers to use

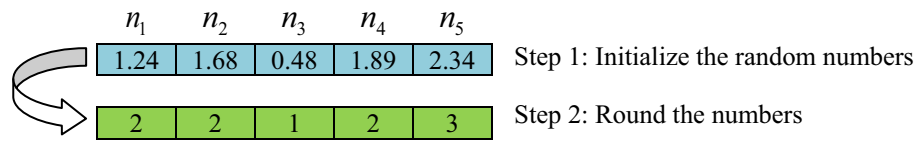


Fig. 6 The used technique to assign a type of vehicle for nurses

this technique to calculate the cost of objective function [34–40].

An example is provided to consider the transformation of solutions. In this regard, consider five nurses (N) with three types of vehicles (K) and twelve patients (M). To implement the plan of representation, firstly, the type of used vehicle for each nurse is should be specified. An array by a length of N is generated by a uniform distribution: $U(0, K)$. After that the type of vehicle assigns to each nurse is computed. These procedures are illustrated in Fig. 6. As indicated, the second type of vehicle is employed by the nurses n_1, n_2 and n_4 . As such, the first and third type of vehicles are utilized for nurses n_3 and n_5 , respectively.

To clarify the route of patients, an array in length of M is distributed by $U(0, 1)$. In the second phase, these numbers are sorted and numbers of patients are determined. Figure 7 illustrates an example of the used arrays of this matrix for algorithms. Then, according to the maximum desired total distance traveled ($MDIS_{nk}$), the capacity of used vehicle (CAP_k) and also the time windows of constraints, the route of four used nurses and patients are specified. In this example, the fifth nurse is not required for assigned nurses. In conclusion, the routes are outputted as follows:

$$n_1 = \{m_2 \rightarrow m_1 \rightarrow m_8\}, \quad n_2 = \{m_5 \rightarrow m_{11} \rightarrow m_3\}, \\ n_3 = \{m_{10} \rightarrow m_9 \rightarrow m_7 \rightarrow m_{12}\}, \quad n_4 = \{m_4 \rightarrow m_6\}$$

4 Computational results

In this section, first of all, the test problems are generated by an approach regarding the developed HHCP. After that the Taguchi method is applied to tune the proper values for algorithms' parameters. A comprehensive analysis is then given by the comparison of proposed heuristics with hybrid metaheuristic as well as the Lagrangian relaxation-based algorithm. The advantages and disadvantages of solution approaches are discussed as well. Finally, the validation of

developed HHCP is examined by a set of sensitivity analyses. Note that the related codes were written in JAVA (NetBeans IDE 8.0.2) on a computer with Intel® Core™ i5-2400 3.10 GHz in CPU and 4 GB in RAM and using Windows 10 as an operating system.

4.1 Data generation

Due to the novelty of proposed model, there are not any available benchmarks in the literature to test the developed HHCP. To do this end, a methodology is required to generate the test problems. Twelve test problems including three classifications, i.e., small: SP1 to SP4, medium: MP5 to MP8 and large sizes: LP9 to LP12 are presented. Table 1 shows the sizes of problem instances. To have a fair the comparison between algorithms, the maximum number of iterations for hybrid VNS-SA metaheuristic as well as Lagrangian relaxation-based algorithm (LB) is set by increasing the size of problem. These numbers are estimated according to previous studies [27, 38].

Notably, the most significant factors to create the HHCP are the transportation cost (TC_k) and their capacities (CAP_k) for used vehicles as can be seen in Table 2. Furthermore, the range of other parameters is given according to Table 3. It should be noted that to employ the generated benchmarks for future works, the source of data is available online as a supplementary material attached to this paper.

4.2 Parameters tuning

To have a fair comparison between the algorithms, a plan for tuning the algorithms' parameters should be examined [25, 26]. If the algorithms are not tuned well, their behavior would be inefficient [42, 43]. Here, Taguchi method offered by Taguchi [44] is utilized to achieve this. This cost-efficient approach has been popularized as a robust parameter design for algorithms. This method was first considered to decrease the number of experiments in

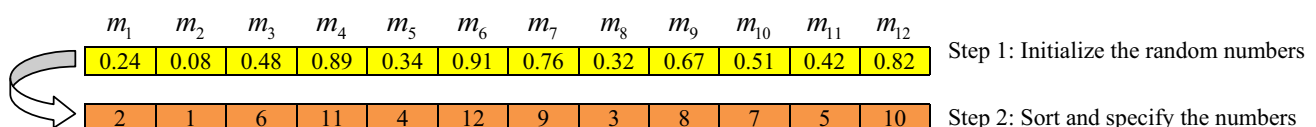


Fig. 7 The used technique to assign patients to each route

Table 1 The instances for test problem

Classification	Instance	Number of nurses (N)	Types of vehicles (K)	Number of patients (M)	Maximum number of iterations	
					VNS-SA	LB
Small	SP1	2	2	10	1000	5
	SP2	3	2	25	1000	5
	SP3	4	3	40	1100	15
	SP4	6	3	65	1200	20
Medium	MP5	8	3	80	1500	50
	MP6	8	4	85	1500	50
	MP7	9	5	95	1700	60
	MP8	10	5	100	1800	70
Large	LP9	12	6	120	2000	80
	LP10	15	6	150	2500	90
	LP11	16	7	160	2600	90
	LP12	20	8	200	3000	100

Table 2 Details of used vehicles in our study

Number	Name of vehicle	TC_k	CAP_k
1	Small car 1 passenger	1	200
2	Small car 2 passengers	2	250
3	Large car 1 passenger	2.5	300
4	Large car 2 passengers	5	500
5	Large car 4 passengers	10	1000
6	Train	50	10,000
7	Coach	200	50,000
8	Plane	1000	100,000

quality engineering topics [43]. In this method, the design factors are divided into two types: control factors and noise factors. Due to unpractical, often impossible omission of the noise factors, this methodology aims to minimize the effect of noise and reaching the best level of the influential controllable factors on the basis of robustness. This method determines the relative importance of each factor with respect to its main impacts on the performance of the algorithm. A transformation of the repetition data to another value is the measure of variation. Generally, there are two well-known methods for the measure of variation including signal-to-noise (S/N) and relative percentage deviation (RPD). For more details, interested readers are referred to the works done by [29, 35–37, 40].

Regarding the S/N ratio, it indicates the amount of variation in response variable of objective function (Z) [37]. The higher value of S/N brings a better quality. For a minimizing mathematical formulation, the following equation is considered to calculate the S/N ratio:

Table 3 Parameters and their surfaces

Parameters	Surfaces
(x_i, y_i)	$1000 \times (U(0, 1), U(0, 1))$
(x_j, y_j)	$1000 \times (U(0, 1), U(0, 1))$
D_{ij}	$\sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$
A_i	$\text{Rand}\{5, 10, 15, \dots, 100\}$
B_i	$A_i \times U(0, 1)$
PEN	For small sizes: 1.5, medium sizes: 3, large sizes: 4.5
W_i	$\text{rand}\{5, 10, \dots, 60\}$
E_i	$\text{rand}\{0, 1, \dots, 10\}$
L_i	$\text{rand}\{500, 1000, \dots, 5000\}$
T_{ij}	$\frac{D_{ij}}{\sum_{i=1}^M \sum_{j=1}^M D_{ij}} \times 120 \times S^a$
$MDIS_{nk}$	$\text{rand}\{25, 000, 30, 000, \dots, 50, 000\}$

^aSpeed transfer coefficient according to the used vehicle for transferring between patient i and j

$$S/N = -10 \times \log_{10}(Z)^2 \quad (19)$$

Each parameter of algorithms in Taguchi method is a factor with certain required levels. Accordingly, this study considers maximum four levels for the factors of LB and maximum three levels for VNS-SA's factors. The complementary information about the factors of two different metaheuristics is given in Table 4. The procedures of neighbor's shaking include swap, insertion and reversion which are taken from [35–37, 40].

Due to of the randomization of the algorithms, each algorithm is run 30 times independently. The RPD for these

Table 4 Algorithms' factors and their levels

Algorithm	Factor	Levels			
		1	2	3	4
VNS-SA	A: Number of neighbor's shaking ($K_{\max}$)	1 ^a	2 ^a	3 ^a	–
	B: Initial temperature (T)	1000	1500	2000	–
	C: Rate of reduction (α)	0.85	0.9	0.99	–
LB	A: Lagrangian multiplier (π)	0.01	0.1	0.5	0.8
	B: Updater of Lagrangian multiplier (f)	1	1.5	1.8	1.95
	C: Rate of reduction (α)	0.85	0.9	0.95	0.99

^a1: swap; 2: swap and insertion; 3: swap, insertion and reversion

30 runs is calculated using the following equation for a minimization mathematical model:

$$RPD = \frac{Alg_{sol} - Min_{sol}}{Min_{sol}} \quad (20)$$

where Min_{sol} is the minimum best solution among all solutions and Alg_{sol} is the output of algorithm. The lower value of RPD brings the better quality.

To decrease the number of trials for both implemented algorithms, Taguchi method suggests a set of orthogonal array. According to the factors and their levels, Taguchi method offers L_9 for VNS-SA as well as L_{16} for LB method. The results of RPD and S/N of algorithms are depicted in Figs. 8, 9, 10 and 11. The proper levels of algorithms are evident from the figures. As a result, the tuned parameters of algorithms are given in Table 5.

4.3 Comparison of algorithms

In this section, the Lagrangian relaxation-based algorithm, three fast heuristics and a hybrid metaheuristic are compared with each other. The metrics to do this comparison are the solution quality and its discrepancy from the best solution ever found, the best, the worst, the average and the standard deviation of algorithms' solutions during thirty run times; hitting time, that is, the first time in which the best solution has been seen and after that an improvement on the solution has not been occurred [35–37, 42], and the computation time of algorithms. Note that the proposed hybrid metaheuristic is divided into four versions. To

validate the results of LB and heuristics along with hybrid algorithms, GAMS software as a preliminary exact method (EX) has been utilized to solve the instances. In the four hybrid algorithms, three of them are based on the initial solution of heuristics in addition to general version of VNS-SA.

The results are given in Table 6. Figure 12 shows the behavior of algorithms in the term of discrepancy to the best solution. The computational time of algorithms is compared in Fig. 13 as well. Similarly, the behavior of hitting time for metaheuristics to assess the convergence rate of algorithms is depicted in Fig. 14. To find out the best technique, the statistical analyses are done in Fig. 15.

Inspecting the results in Table 6, the proposed Lagrangian relaxation-based algorithm finds a feasible LB in half of the test problems (SP1 to MP6). Note that the EX has been solved only SP1 to MP5 test problems. The outputs of exact software reveal that LB finds the global solutions for the test problems of SP1 to MP6. Although the exact method cannot available for the medium and large instances, the LB method can handle the large instances as well. In the rest of the test problems, a feasible LB has not reached until the end of iterations. In regard to heuristics, the output of algorithms and their computational time along with the discrepancy to the LB solution are noticeable. To have comprehensive comparison among four hybrid algorithms proposed, the best, the average and the worst along with the standard deviation of solutions during thirty run times, the computational and hitting time of algorithms and

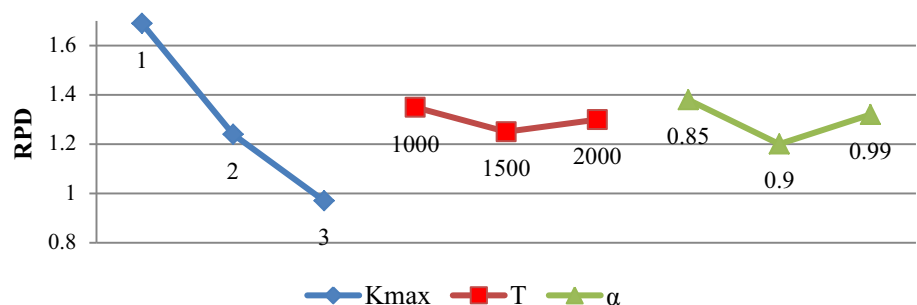
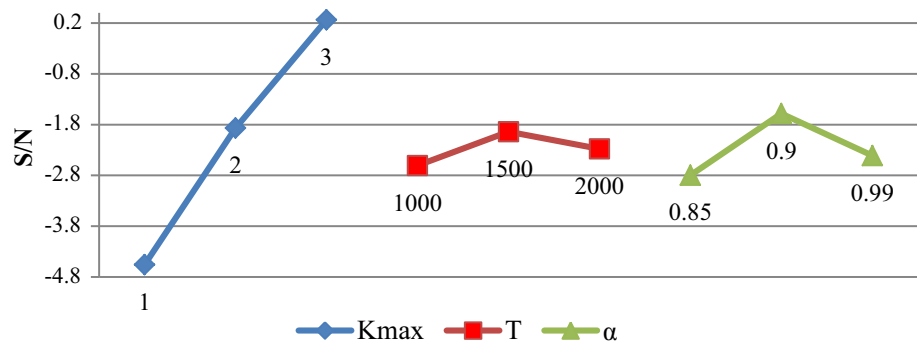
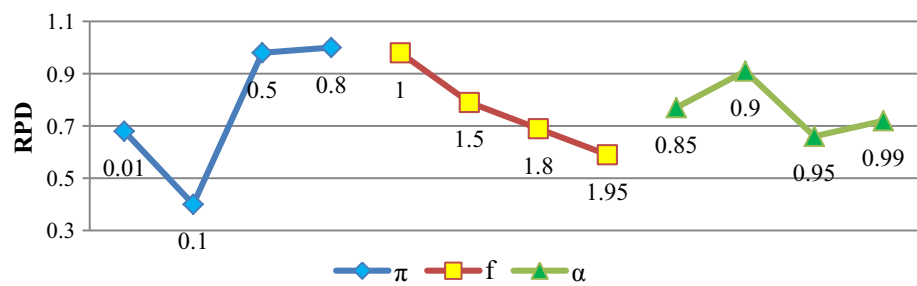
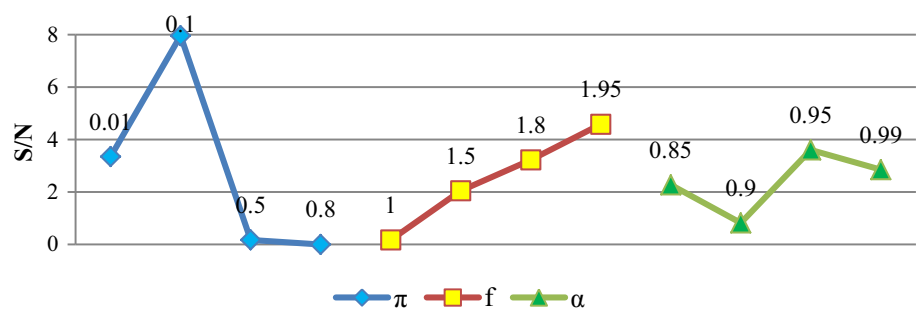
Fig. 8 Average RPD for VNS-SA's parameters

Fig. 9 S/N ratio for VNS-SA's parameters**Fig. 10** Average RPD for LB's parameters**Fig. 11** S/N ratio for LB's parameters**Table 5** Calibrated parameters of algorithms

Algorithm	Parameters
VNS-SA	$K_{\max} = 3$; $T = 1500$; $\alpha = 0.9$
LB	$\pi = 0.1$; $f = 1.95$; $\alpha = 0.95$

also the discrepancy of best solution of metaheuristics from the solution outputted from LB are also evident in Table 6.

As indicated from Fig. 12, it is seen that heuristics provide a higher discrepancy in comparison with metaheuristics. Note that especially in medium sizes and large sizes, heuristics are mostly better than VNS-SA. Overall, H3 shows a weak behavior in this item except in SP3, SP4, MP5, MP8, LP11 and LP12 in which H1 is the worst algorithm. In contrast, H2 is the best heuristic among heuristics. By another point of view, the hybrid metaheuristics utilizing the initial solutions of heuristics are more powerful than their original algorithm. Among them,

the VNS-SA-H2 is the best technique in this item and reaches the best optimal solution in this comparison.

As per the results in Fig. 13, it is evident that the LB requires more computational time. The main reason is due to the randomization of UP generating from the structure of algorithm along with increasing the maximum number of iteration. As such, similar pattern is observed for the metaheuristics employed. While the computational time of metaheuristics is significantly lower than LB method, they are slower than heuristics.

What can be seen at the first glance of Fig. 14 is that the hitting time of hybrid of VNS-SA and heuristics is better than VNS-SA in general. It may be resulted that the initial solutions of heuristics help the metaheuristic to have a better convergence rate. Furthermore, the graph reveals a similarity between VNS-SA and VNS-SA-H3 as well as between VNS-SA-H1 and VNS-SA-H2. Accordingly, VNS-SA-H3 is mostly better than VNS-SA in overall except in a few test problems, i.e., SP4 and MP5. Another fact is that VNS-SA-H2 is clearly better than VNS-SA-H1

Table 6 The results of comparison (OUT = the average of solutions; CPU = the computational time-based second; GAP = the deviation of algorithms from LB; B = the best; W = the worst; SD = the standard deviation, HT = the hitting time based on second)

Algorithm	SP1	SP2	SP3	SP4	MP5	MP6	MP7	MP8	LP9	LP10	LP11	LP12
EX	OUT	3255	5033	5433	7264	7751	–	–	–	–	–	–
LB	OUT	3255	5033	5433	7264	7751	7988	8952	9548	9986	10,609	10,954
	Feasibility	✓	✓	✓	✓	✓	✓	–	–	–	–	–
H1	CPU	10.86	12.54	25.71	78.42	162.67	186.88	481.64	768.89	1367.9	1712	1763
	OUT	3904.2	5246.8	5637	8912.5	8894.9	9297.2	9060.6	10,528	10,223	11,516	12,762
	CPU	0.0404	0.0418	0.0664	0.0430	0.0761	0.1912	0.34407	0.5153	0.6665	0.755	0.8273
	GAP	0.1994	0.0424	0.0375	0.2269	0.1475	0.1638	0.012131	0.1026	0.0237	0.0854	0.16505
H2	OUT	3643.3	5158.2	5479.3	8538.1	8205.4	8180.7	9126.8	9992.3	10,707	11,709	12,494
	CPU	0.0476	0.0423	0.0344	0.0454	0.0806	0.0736	0.0893	0.1066	0.1157	0.2094	0.2456
	GAP	0.1192	0.0248	0.0085	0.1753	0.0586	0.0241	0.01952	0.0465	0.0722	0.1036	0.1405
H3	OUT	4168.2	5574.1	5538.2	8851.7	8726.4	9477.1	9735.2	10,224.5	11,168	12,675	12,067
	CPU	0.044	0.04205	0.0504	0.0442	0.07835	0.1324	0.216685	0.31095	0.3911	0.4822	0.5364
	GAP	0.2805	0.1075	0.0193	0.21857	0.12584	0.18641	0.08748	0.07085	0.1183	0.1947	0.1016
VNS-SA	B	3255	5033	5578	7764	7988	8365	9060.6	10,174	10,582	11,334	12,366
	OUT	3457.8	5186.9	5699	7964	8256	8652	9185.2	10,305	10,836	11,586	12,724
	W	3705	5438	5875	8266	8697	9035	9366	10,412	10,945	11,925	12,863
	SD	123.95	122.66	97.075	176.9	268.5	268.91	130.52	107.28	176.95	296.56	269.23
	HT	6.88	8.75	14.82	22.89	28.65	30.92	33.16	41.73	65.82	71.39	95.64
	CPU	8.56	12.86	16.37	29.67	34.82	42.15	45.83	56.91	78.91	84.11	109.38
	GAP	0	0	0.0266	0.0688	0.0305	0.0471	0.0121	0.0655	0.0596	0.0683	0.1289
VNS-SA-H1	B	3255	5033	5433	7673	7925	8177	9060.6	9877.4	10,178	11,071	11,942
	OUT	3440	5097	5482	7837	8155	8469	9060.6	9938.5	10,189	11,205	12,394
	W	3682	5175	5586	8219	8526	8871	9060.6	10,326	10,215	11,487	12,562
	SD	117.77	42.668	50.784	196.11	227.43	278.75	0	218.95	18.05	212.34	336.69
	HT	5.68	6.73	6.89	17.83	20.99	19.54	0.34407	18.83	15.47	23.14	64.18
	CPU	9.08	10.87	17.54	28.59	37.22	45.72	49.11	59.42	88.21	93.15	118.42
	GAP	0	0	0	0.0563	0.0224	0.0236	0.0121	0.0344	0.0192	0.0435	0.0901
VNS-SA-H2	B	3255	5033	5433	7369	7846	7988	9060.6	9764.3	10,178	10,994	11,755
	OUT	3419	5099.4	5446.2	7748.3	7908.1	8026	9068.5	9782.5	10,344	11,267	11,984.2
	W	3574	5126.4	5471.5	8305	8098.2	8073	9089.2	9854.3	10,682	11,543	12,289.5
	SD	87.73	28.839	12.716	329.5	98.55	34.06	12.55	42.827	244	274.5	281.5
	HT	4.92	5.18	2.54	15.82	24.83	14.39	4.17	7.62	20.65	38.33	44.81
	CPU	8.94	11.56	18.35	29.42	38.51	46.82	50.42	57.13	85.91	94.26	113.27
	GAP	0	0	0	0.0144	0.0122	0	0.0121	0.0226	0.0192	0.0362	0.0731
												0.0603

Table 6 (continued)

Algorithm	SP1	SP2	SP3	SP4	MP5	MP6	MP7	MP8	LP9	LP10	LP11	LP12
VNS-SA-H3	B	3255	5033	5433	7488	7925	8244	9060.6	9877.4	10,178	11,071	11,862
	OUT	3518.7	5234.8	5465.7	8101.7	8279.7	8804.7	9298.9	10,037.9	10,630	11,734.5	11,900
	W	3782.5	5436.7	5498.5	8715.4	8634.5	9365.4	9537.2	10,198.5	11,082	12,398	11,938
	SD	145.06	121.1	21.28	429.59	266.06	448.56	202.55	144.49	429.4	663.5	39.9
	HT	5.89	9.53	6.72	26.88	32.79	31.75	25.15	43.86	55.72	47.61	65.19
	CPU	9.52	12.85	19.33	30.91	40.16	49.57	51.22	59.84	87.19	92.32	115.62
	GAP	0	0	0	0.0308	0.0224	0.03204	0.0121	0.0344	0.0192	0.0435	0.0828
												0.0683

The best values are shown in bold

especially in small and medium sizes. Conversely, VNS-SA-H1 shows a better convergence rate in a few large-size problems. As a result, the VNS-SA-H2 is the best technique among other similar ones in this item.

To choose the best technique, a set of statistical analyses is conducted here. To do this end, the standard deviation of algorithms due to their stochastic nature of metaheuristics is transformed to a popular metric to make the results reliable. Therefore, the relative deviation index (RDI) has been formulated as follows:

$$RDI = \frac{|Alg_{sol} - Best_{sol}|}{Max_{sol} - Min_{sol}} \quad (21)$$

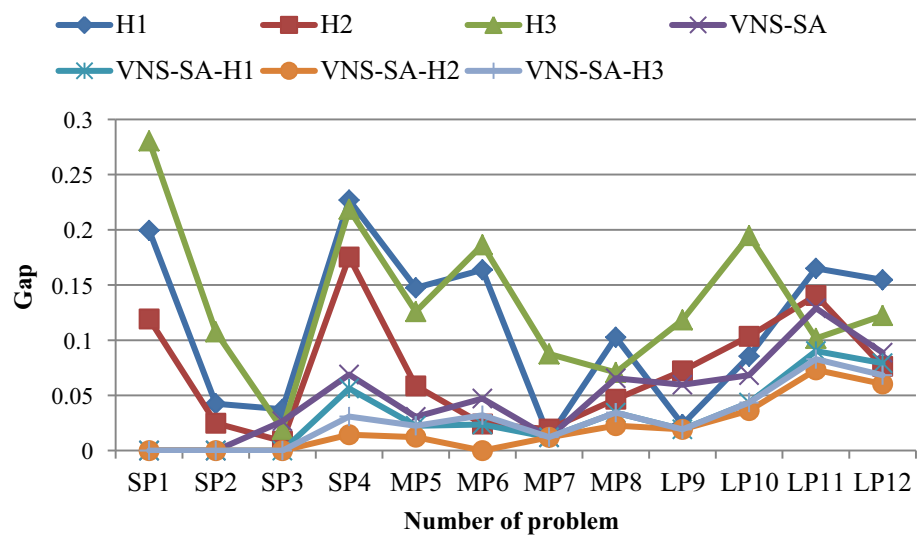
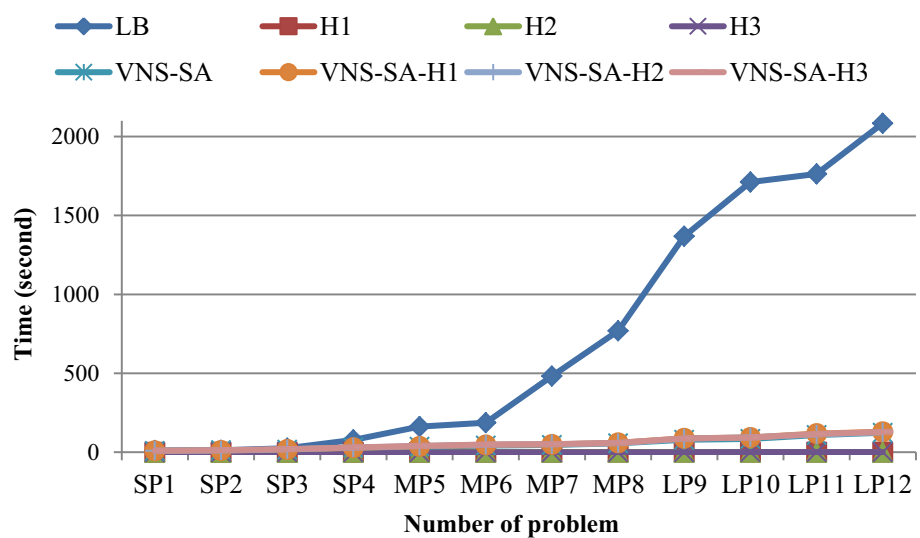
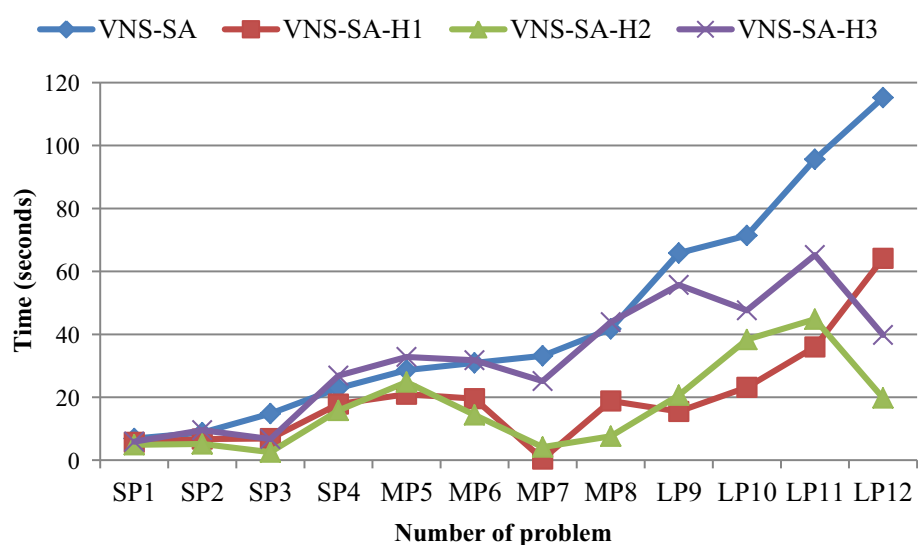
where Alg_{sol} is the objective value obtained by an algorithm, Max_{sol} and Min_{sol} are, respectively, the maximum and the minimum values among all values outputted by algorithms. The lower value of RDI , the better algorithm is [25, 26]. The means plots and least significant difference (LSD) for the proposed hybrid algorithms are provided in Fig. 15. This figure is divided into three graphs to consider the best algorithm in small, medium and large sizes, separately.

From Fig. 15a, there is not a clear difference between VNS-SA and VNS-SA-H3 which are the worst algorithms for small-size case studies. It is evident that they have similar values of mean and LSD. Similarly, the performance of VNS-SA-H1 and VNS-SA-H2 is the same in overall. As a result, while the VNS-SA is the worst algorithm, VNS-SA-H2 is the best one in this case.

Regarding the medium-size problems (Fig. 15b), there is a clear difference between VNS-SA-H2 and other metaheuristics. It is evident that VNS-SA-H2 is the best algorithm in this case as well. On the other hand, the robustness of VNS-SA is slightly better than two other heuristic-based algorithms, *i.e.*, VNS-SA-H1 and VNS-SA-H3.

In addition to the medium- and small-size, the similarities of algorithms in large-size case study have been increased in comparison with the other case studies. It seems that it is difficult to select the best and the worst algorithm. However, the robustness of VNS-SA-H2 is generally better than other algorithms in the large sizes (Fig. 15c). Finally, it can be resulted from the experiments done that the VNS-SA-H2 is the best technique in this study to better solve the proposed HHCP.

In conclusion, there are some advantages and disadvantages regarding the proposed heuristics. The main advantage of second heuristic is to have less computational time and better reach an optimal solution. Using this heuristic can improve the performance of metaheuristics. Note that the main disadvantage of proposed LB is its computational cost and not be able to solve the large

Fig. 12 The comparison of gap behavior of algorithms**Fig. 13** The behavior of algorithms in term of computational time**Fig. 14** The comparison of hitting time of hybrid constructive metaheuristic algorithms

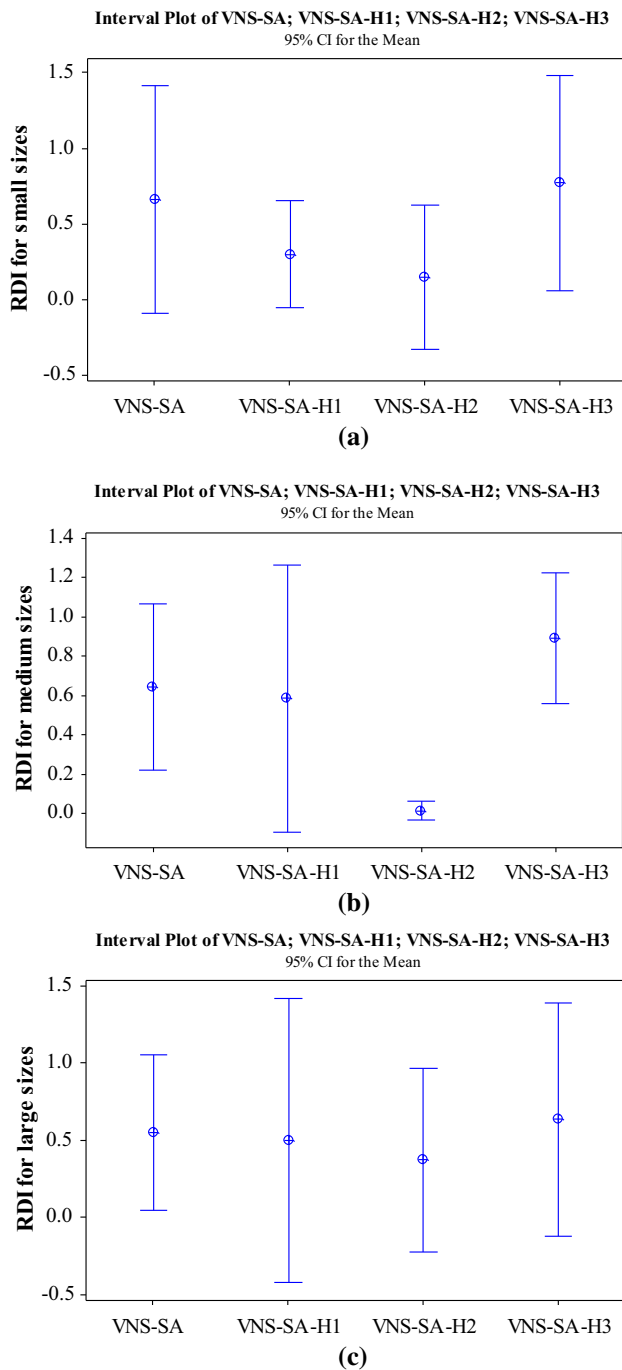


Fig. 15 Interval plot of hybrid constructive metaheuristics in small (a), medium (b) and large (c) sizes

instances. Therefore, more efforts may be needed to improve its limitations.

4.4 Sensitivity analyses

To verify the properties of developed HHCP, a set of sensitivity analyses have been performed on the critical parameters. A small test problem, e.g., SP3 considering

four nurses, three types of vehicles and 40 patients, is chosen. To address the problem, the Lagrangian relaxation-based algorithm reaching a feasible lower bound is utilized. The main parameters of proposed HHCP are including the effects of the penalty value for overall traveled distance of the length of the tour of nurses to visit the patients (PEN), the number of nurses (N), the number of distributed patients (M) and the types of employed vehicles (K). Each treatment is divided into four cases, i.e., C1 to C4 to consider the behavior of objective model. Overall, the related outputs are provided in Tables 7, 8, 9 and 10 and also Fig. 16, 17, 18 and 19.

The main innovation of model is to consider a policy to control the overall distance. Accordingly, the penalty value of the overall traveled distance for the length of tour of nurses to visit their patients (PEN), and sensitivity analyses have been performed by increasing the amount of this parameter. Table 7 provides the details of experiments. To identify the behavior of objective function, the given values are depicted in Fig. 16. Inspecting the results, generally an increase in this parameter leads to an increase in the behavior of objective function. Although the growth rate of objective function by increasing the parameter is decreased by the end of cases, this rate uniformly is increased as the same. As a result, it seems that the higher value of this parameter can reduce the total cost more efficiently.

The number of employed nurses (N) is considered to perform a number of analyses. The results are given in Table 8. To identify the behavior of objective function during the changes, Fig. 17 shows the analyses. By increasing the number of nurses, the behavior of the total cost is increased uniformly. In two last cases, the behavior is fixed. Accordingly, the number of tours to visit the patients has a similar behavior. Because of these, additional nurses are not available to satisfy the length of tours to visit all patients.

According to the number of distributed patients (M), the complexity of problem was varied as denoted by previous experiments. The sensitivity analyses on the number of distributed patients have been done, and the results are reported in Table 9. The behavior of objective function is depicted in Fig. 18. It is evident that by increasing the

Table 7 Sensitivity analyses on the penalty value for overall traveled distance of the tour of nurses

Number of cases	PEN	Z
C1	1.5	10,959
C2	2.5	12,292.33
C3	3.5	13,625.67
C4	4.5	14,959

Table 8 Sensitivity analyses on the number of employed nurses

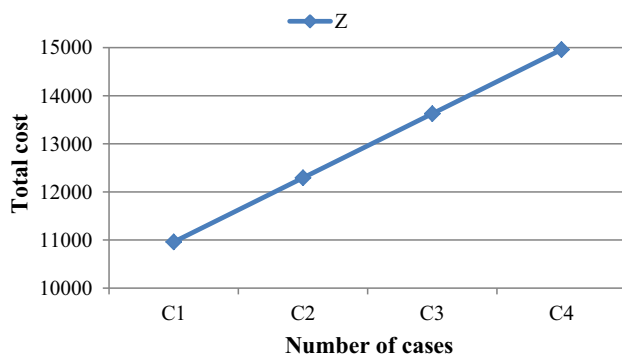
Number of cases	N	Z
C1	3	9875
C2	4	10,959
C3	5	11,576.28
C4	6	11,576.28

Table 9 Sensitivity analyses on the number of distributed patients

Number of cases	M	Z
C1	30	10,546.837
C2	40	10,959
C3	50	14,876
C4	60	17,076

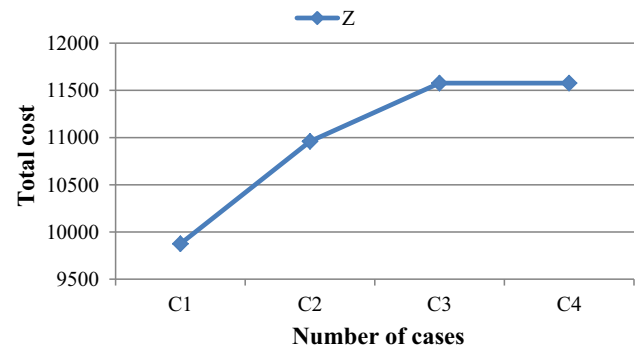
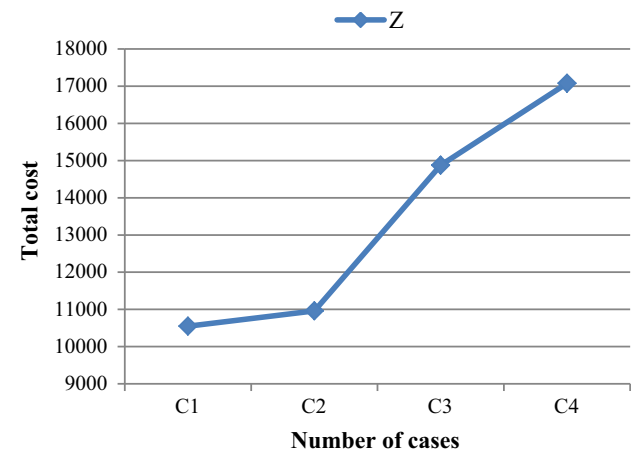
Table 10 Sensitivity analyses on the number of types of employed vehicles

Number of cases	K	Z
C1	3	10,959
C2	4	11,759
C3	5	17,876
C4	6	19,076

**Fig. 16** Behavior of objective function for the sensitivity analyses on the overall traveled distance for the length of tours of nurses

number of distributed patients, the objective values increase. To consider the details of analyses, note that the growth rate of objective value increases slowly, after that, it increases exponentially. Generally, the results show that the sensitivity of number of distributed patients is very high on the objective value.

Another main innovation of this study is to consider the different types of vehicles. This diversification can help the managers to select the best strategy for the total cost. Accordingly, a number of analyses on the number of types

**Fig. 17** Behavior of objective function for the sensitivity analyses on the number of employed nurses**Fig. 18** Behavior of objective function for the sensitivity analyses on the number of distributed patients

of vehicles (K) have been performed by increasing the amount of this parameter, uniformly. Table 10 and Fig. 19 show the description of these analyses. Similar to the last analyses, the results show the high effect of different number of vehicles on the objective functions. Generally, the total cost is increased exponentially. Firstly, the growth rate is slow. Then, it increases so fast. In regard to the managerial insights of the proposed HHCP, selecting the best strategy to employ the most performance transforming systems is so significant to manage efficiently the whole of HHC organizations.

5 Conclusion and future remarks

This study proposed a new formulation for the HHCP considering a policy for overall distance by different type of vehicles. The main innovations were offered to solve this problem by not only a lower bound based on a

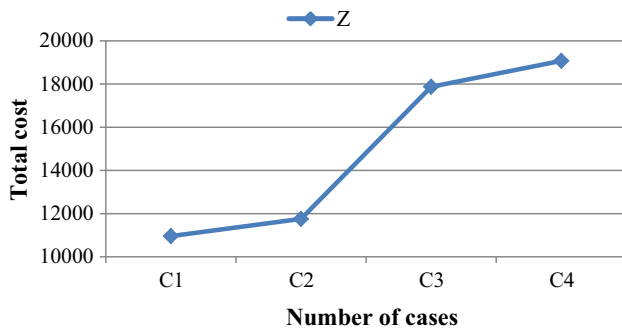


Fig. 19 Behavior of objective function for the sensitivity analyses on the number of types of used vehicles

Lagrangian relaxation-based algorithm but also three fast heuristics along with a hybrid constructive metaheuristic. The algorithms were tuned by the Taguchi experiments to set a proper value for parameters. Finally, the comparisons with algorithms as well as some sensitivity analyses on the key parameters of model were performed.

The results showed that the proposed Lagrangian relaxation method can find a feasible lower bound in small- and medium-size case studies. However, finding a feasible lower bound in large sizes is not possible in this study. The main disadvantage of this algorithm is the higher computational cost as compared to heuristics and metaheuristics. It was also observed that the heuristics can reach the optimal solution faster than metaheuristic. But, the discrepancy from the best solution may be increased. It is generally more than metaheuristic in some cases. Among heuristics, the second version (H2) was the best. Similarly, the hybrid algorithm called VNS-SA-H2 showed the best results and is the best technique in this study. According to the sensitivity analyses, the importance of the penalty value for overall distance as well as different types of vehicles and the others were discussed and analyzed. As a result, the main managerial insights based on the results can be ordered as follows:

- The HHCP is so practical, and one of hot topics in the healthcare systems and its related decisions are operational which should be taken in less time. Accordingly, the proposed heuristics as well as metaheuristics are useful for managers of such systems.
- If the decisions of HHC organization are strategic and the solution quality is very important, an exact solver is required in this case. Therefore, the validation of heuristics and metaheuristics should be examined by a Lagrangian relaxation-based algorithm like this study.
- Controlling the transportation cost as the main cost of HHC organization should be performed by setting a number of parameters to sort the patients similar to the proposed formulation in this study.

- Different types of vehicles are so effective on the cost of transportation cost. As such, the availability time of patients as well as the capacity of vehicles as the executive constraints should be analyzed more by managers of HHCP.

This study opens several new avenues for future works. First of all, more in-depth analyses on the proposed Lagrangian relaxation-based algorithm as well as proposed heuristics can be done. New heuristics and metaheuristics can be examined by generated benchmarks in future study. The proposed mathematical formulation may be expanded by some other new suppositions in this research area. For example, green and sustainable dimensions can be added into the formulation by adding new objective functions to transform it to a multi-objective model. Considering multi-depot of proposed HHCP by using a set of pharmacies and laboratories is another interesting continuation of this study. The patients' inconvenience may be suggested to add into the formulation. Considering the cross-docking operations into the constraint is recommended as well.

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